

# A QUANTIFIER OBSTRUCTION IN AN SDP TEST FOR THE PPT-SQUARED CONJECTURE

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**ABSTRACT.** The PPT-squared conjecture asks whether the composition of two positive-partial-transpose quantum channels is necessarily entanglement breaking. We audit a recently proposed semidefinite-programming test for counterexamples. Its displayed optimization minimizes the trace-norm distance of a composed Choi matrix from the cone of decomposable matrices. We prove that this minimum is identically zero for every fixed positive map. The zero map gives the obstruction in the unnormalized cone, and replacer channels show that imposing trace preservation does not repair it. We give the correctly quantified maximization test, derive an exact dual formula for the decomposability distance, and obtain a bilinear formulation suitable for certified search. We also audit the released implementation and report a reproducible dimension-four search over several structured families and 440 corrected max-dual starts. No exact counterexample was found. Thus the paper corrects a computational method but does not resolve the dimension-four PPT-squared conjecture.

## 1. INTRODUCTION

Let  $\Phi, \Psi : \mathbb{M}_d \rightarrow \mathbb{M}_d$  be quantum channels. A channel is called *PPT* when it is both completely positive and completely copositive. The PPT-squared conjecture asserts that  $\Psi \circ \Phi$  is entanglement breaking whenever both factors are PPT. The problem was formulated in the wake of a 2012 BIRS workshop and admits several equivalent operator-algebraic forms; see Christandl, Müller-Hermes, and Wolf [3]. The complex dimension-three case was proved independently in [3, 2], while the unrestricted problem in dimension four remains open [1]. Recent work proves special symplectic covariance classes [5]; a differentiable-SDP study of a rectangular positive-map/PPT-map formulation found no violation in its tested regime, without constituting a proof [4].

Vom Ende, Khatri, and Denisov [6, Sec. 5] proposed an optimization test based on the distance of a Hermitian Choi matrix from the cone of decomposable matrices. For a fixed positive map  $P$ , their test has the schematic form

$$(1) \quad \min_{T \in \text{CP} \cap \text{coCP}} D(J(P \circ T)),$$

and a positive minimum was interpreted as being equivalent to the existence of a counterexample generated by  $P$ .

The main point of this paper is an elementary but decisive quantifier obstruction: the minimum in (1) is always zero. This is not merely a missing normalization. Even after restricting  $T$  to PPT trace-preserving channels, replacer channels force the same conclusion. The existential search must use a *maximum*. That correction changes the problem from one convex semidefinite program into convex-function maximization, or equivalently into a bilinear conic optimization.

Our contributions are as follows.

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- (i) We give exact proofs that both the unnormalized and trace-preserving versions of the proposed minimum vanish identically.
- (ii) We state the correct compact maximization criterion and derive a trace-norm dual formula that leads to a proof-carrying bilinear search.
- (iii) We verify that the released implementation contains the same zero solution, because it imposes neither trace nor partial-trace normalization.
- (iv) We report a reproducible dimension-four audit. It found no exact counterexample, and we explicitly separate this numerical outcome from the proved correction.

## 2. PRELIMINARIES

We use the Choi convention

$$J(\Phi) = \sum_{i,j=1}^d E_{ij} \otimes \Phi(E_{ij}).$$

Then  $\Phi$  is completely positive if and only if  $J(\Phi) \succeq 0$ , and it is completely copositive if and only if  $J(\Phi)^\Gamma \succeq 0$ , where  $\Gamma$  denotes partial transpose on the second tensor factor. Trace preservation is equivalent to  $\text{Tr}_2 J(\Phi) = I_d$ . Consequently, the Choi matrix of a channel on a four-dimensional system has size  $16 \times 16$ .

Let

$$\text{Dec}_d = \{A + B^\Gamma : A, B \succeq 0\}$$

be the cone of decomposable Hermitian matrices. Following [6], define

$$(2) \quad D(W) = \inf_{A, B \succeq 0} \|W - A - B^\Gamma\|_1 = \text{dist}_{\|\cdot\|_1}(W, \text{Dec}_d).$$

The decomposable cone is closed, so  $D(W) = 0$  exactly when  $W \in \text{Dec}_d$ . In particular  $D(W) \geq 0$ .

For a fixed positive map  $P : \mathbb{M}_d \rightarrow \mathbb{M}_d$ , let

$$(3) \quad \mathcal{K}_{\text{PPT}, \text{TP}} = \{J(T) : J(T) \succeq 0, J(T)^\Gamma \succeq 0, \text{Tr}_2 J(T) = I_d\}.$$

This is a nonempty compact spectrahedron.

## 3. THE QUANTIFIER OBSTRUCTION

**Theorem 3.1** (Unnormalized obstruction). *For every positive map  $P : \mathbb{M}_d \rightarrow \mathbb{M}_d$ ,*

$$\min_{T \in \text{CP} \cap \text{coCP}} D(J(P \circ T)) = 0.$$

*Proof.* The zero map belongs to  $\text{CP} \cap \text{coCP}$ . Hence the displayed minimum is at most

$$D(J(P \circ 0)) = D(0) = 0.$$

Nonnegativity of  $D$  gives the reverse inequality.  $\square$

It is natural to try to remove the zero map by imposing trace preservation. The next result shows that this does not fix the minimization.

**Theorem 3.2** (Trace-preserving obstruction). *For every positive map  $P : \mathbb{M}_d \rightarrow \mathbb{M}_d$ ,*

$$\min_{J(T) \in \mathcal{K}_{\text{PPT}, \text{TP}}} D(J(P \circ T)) = 0.$$

*Proof.* Let  $\tau \succeq 0$  with  $\text{Tr} \tau = 1$ , and define the replacer channel

$$T_\tau(X) = \text{Tr}(X)\tau.$$

Its Choi matrix is  $I_d \otimes \tau$ . It is separable and therefore positive under partial transpose; moreover  $\text{Tr}_2(I_d \otimes \tau) = I_d$ . Thus  $J(T_\tau) \in \mathcal{K}_{\text{PPT}, \text{TP}}$ .

Positivity of  $P$  gives  $P(\tau) \succeq 0$ , and

$$(P \circ T_\tau)(X) = \text{Tr}(X)P(\tau).$$

This is a completely positive measure-and-prepare map, with Choi matrix  $I_d \otimes P(\tau) \succeq 0$ . Its Choi matrix lies in  $\text{Dec}_d$ , so  $D(J(P \circ T_\tau)) = 0$ . Nonnegativity of  $D$  completes the proof.  $\square$

**Remark 3.3** (The logical issue). The desired statement is existential: there should be *some* feasible  $T$  for which the distance is positive. A minimum instead asks whether *every* feasible  $T$  has positive distance. Theorems 3.1 and 3.2 exhibit canonical feasible maps at distance zero and therefore rule out this stronger universal condition.

#### 4. THE CORRECTLY QUANTIFIED OPTIMIZATION

Compactness of (3) immediately gives the proper fixed- $P$  test.

**Corollary 4.1** (Correct existence criterion). *For every positive map  $P : \mathbb{M}_d \rightarrow \mathbb{M}_d$ ,*

$$\exists J(T) \in \mathcal{K}_{\text{PPT,TP}} : D(J(P \circ T)) > 0 \iff \max_{J(T) \in \mathcal{K}_{\text{PPT,TP}}} D(J(P \circ T)) > 0.$$

*The maximum is attained.*

The objective is convex in  $J(T)$ , so Corollary 4.1 is a convex-function maximization problem rather than a convex SDP. An exact dual representation makes the nonconvexity especially transparent.

**Proposition 4.2** (Dual decomposability distance). *For every Hermitian  $W$ ,*

$$(4) \quad D(W) = \max\{\text{Tr}(YW) : -I \preceq Y \preceq 0, Y^\Gamma \preceq 0\}.$$

*Proof.* For a closed convex cone  $C$  in a finite-dimensional normed space, the distance from  $W$  to  $C$  is

$$\max\{\langle Y, W \rangle : Y \in C^\circ, \|Y\|_* \leq 1\},$$

where  $C^\circ$  is the polar cone and  $\|\cdot\|_*$  is the dual norm. This follows from finite-dimensional Hahn–Banach separation, or equivalently from Fenchel duality applied to the norm and the indicator of  $C$ .

Here the dual of the trace norm is the operator norm. Since  $\text{Dec}_d = \{A + B^\Gamma : A, B \succeq 0\}$ , its polar cone is

$$\text{Dec}_d^\circ = \{Y : Y \preceq 0, Y^\Gamma \preceq 0\}.$$

For  $Y \preceq 0$ , the condition  $\|Y\|_\infty \leq 1$  is equivalent to  $-I \preceq Y \preceq 0$ . Substitution gives (4). The feasible set is compact, so the maximum is attained.  $\square$

Combining Corollary 4.1 with Proposition 4.2 yields

$$(5) \quad \max_{\substack{J(T) \in \mathcal{K}_{\text{PPT,TP}} \\ -I \preceq Y \preceq 0, Y^\Gamma \preceq 0}} \text{Tr}(YJ(P \circ T)).$$

Each block separately defines an SDP, but the joint objective is bilinear. This supports alternating maximization and branch-and-bound strategies while preventing the incorrect interpretation of a single minimization SDP.

## 5. AUDIT OF THE RELEASED IMPLEMENTATION

The ancillary code accompanying [6] can be audited directly. The routine

`check_PPT_squared_opt`

constrains the variable representing  $J(T)$  only by

$$J(T) \succeq 0, \quad J(T)^\Gamma \succeq 0.$$

There is no trace or partial-trace normalization. Therefore the exact all-zero assignment is feasible, exactly as in Theorem 3.1. Any reported nonpositive optimum from that implementation is consequently uninformative for the existential PPT-squared question.

This observation is deliberately narrow. It does not affect the other results of [6] concerning positive maps. It invalidates the stated PPT-squared existence test and the interpretation of numerical output from that test.

## 6. DIMENSION-FOUR COMPUTATIONAL AUDIT

We implemented both the primal distance (2) and the dual formula (4), with two independent Choi-composition routines. The corrected search uses the bilinear formulation (5). Table 1 records the completed searches. Every entry is a discovery computation, not a proof of separability or of the PPT-squared conjecture.

TABLE 1. Dimension-four search audit. CCNR denotes the computable cross-norm/realignment criterion.

| Route                                             | Coverage                 | Best or terminal observation     |
|---------------------------------------------------|--------------------------|----------------------------------|
| $4 \rightarrow 2 \rightarrow 4$ structured family | 10,000 grid points       | CCNR ratio 0.832996              |
| Rotated GenTiles1 UPB pairs                       | 20,000 pairs             | CCNR ratio 0.521516              |
| Alternating CCNR SDP                              | 33 starts                | CCNR ratio 0.9999538672          |
| Direct witness seesaws                            | 65 starts                | no robust negative witness value |
| Correct max-dual search                           | 440 starts               | no robust positive distance      |
| UPB target factorization                          | 2 starts, 5 alternations | Frobenius residual 0.614201      |

The 440 max-dual starts covered Breuer–Hall, UPB-derived, and embedded Choi positive maps, real and complex variables, and both channel and extreme-dual initializations. One embedded-Choi run initially returned  $2.52 \times 10^{-7}$ , but the candidate violated the PSD/PPT constraints by  $1.62 \times 10^{-7}$  and trace preservation by  $3.91 \times 10^{-7}$ . Projection to the feasible set changed the recomputed dual distance to  $-3.76 \times 10^{-10}$ . We therefore reject, preserve, and label this point as a solver artifact rather than a counterexample.

The test suite contains 17 unit tests covering Choi conventions, direct and superoperator composition, partial transpose, trace preservation, real and complex linear-factor operators, UPB constructions, standard positive maps, and primal–dual distance agreement. Representative structured points were also tested by numerical two-copy PPT symmetric extensions. These checks increase reproducibility but cannot certify the universal conjecture.

## 7. DISCUSSION AND LIMITATIONS

Theorems 3.1 and 3.2, the corrected criterion in Corollary 4.1, and the dual identity in Proposition 4.2 are exact finite-dimensional arguments. They do not depend on floating-point computation. The source-code observation is verified directly from the released constraints.

By contrast, the dimension-four search is numerical. Failure to find a counterexample in finitely many families or local starts neither proves that all compositions are entanglement breaking nor supplies a separability certificate for every composition. No Lean formalization is claimed, and no exact four-dimensional counterexample is claimed. The mathematical status after this work is therefore precise: the proposed minimization test is corrected, while the general dimension-four PPT-squared conjecture remains open.

#### COMPUTATIONAL AND AI DISCLOSURE

Computational tools and an AI coding assistant were used to design and run the search routes, implement independent consistency checks, audit the released SDP, and assist with manuscript preparation. Every theorem in this paper was independently reduced to the explicit arguments printed above. Numerical experiments are reported with their limitations and are not used as proof of the conjecture.

#### DATA AND CODE AVAILABILITY

The reproducibility package accompanying this manuscript contains the search scripts, fixed random seeds, machine-readable result files, rejected-candidate artifacts, 17 unit tests, the audited upstream source, and SHA-256 identifiers for the supplied PDF and source archive.

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